Basics of identification theory

Lecture 4. Identifiability and Stable Least Squares

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Outline

1. **The friction experiment and identifiability** S01–S07
   What do its measurements determine, and how does speed variation change the answer?

2. **Singular values and sensitivity** S08–S12
   Which parameter directions amplify measurement and numerical errors?

3. **Ridge as a modified estimation problem** S13–S16
   Why is its solution unique, and how does it act on each direction?

4. **Bias, variance, and the choice of remedy** S17–S21
   What is gained, what is lost, and when should the experiment change?

Appendix A1–A11 contains proofs, computational examples, and focused reading.

1. BLUE does not mean a small estimation error

From Gauss–Markov, with a fixed matrix $X \in \mathbb{R}^{N\times m}$ of full column rank,

$$Y = X\theta_0 + \varepsilon, \qquad \mathbb{E}\varepsilon = 0, \qquad \mathrm{Cov}(\varepsilon) = \sigma^2 I_N,$$

$$\widehat{\theta}_{\mathrm{LS}} = (X^T X)^{-1} X^T Y, \qquad \mathrm{Cov}(\widehat{\theta}_{\mathrm{LS}}) = \sigma^2 (X^T X)^{-1}.$$

OLS has the smallest covariance *among linear unbiased estimators* for this experiment. The covariance itself can still be very large.

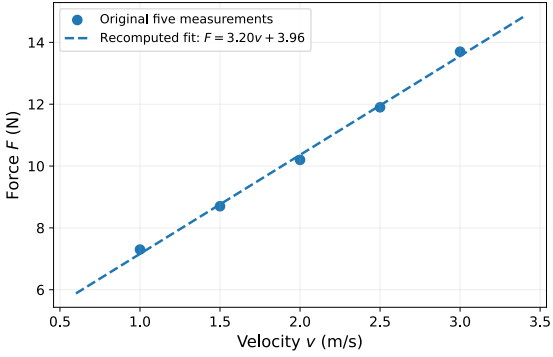
Our question

When do the data determine the parameters uniquely and reliably? What can we change when they do not?

Route: a friction experiment $\longrightarrow$ rank and singular values $\longrightarrow$ sensitivity $\longrightarrow$ ridge and its bias cost.

Prerequisite: Lecture 3, Gauss–Markov. Original source: Lecture 2 PDF, pp. 47–55 [S0].

2. Keep the friction model; change the experiment



Five measurements from the earlier lecture, redrawn together with their least-squares fit.

For positive speed, use the *magnitude* of friction, or the force needed to balance it:

$$F = F_v v + F_c.$$

Fix $v_{\mathrm{ref}} = 1\,\mathrm{m/s}$ and $F_{\mathrm{ref}} = 1\,\mathrm{N}$. Define

$$u = \frac{v}{v_{\mathrm{ref}}}, \quad y = \frac{F}{F_{\mathrm{ref}}}, \quad \theta = \begin{bmatrix} F_v v_{\mathrm{ref}}/F_{\mathrm{ref}} \\ F_c/F_{\mathrm{ref}} \end{bmatrix}.$$

Then $y = [\,u \;\; 1\,]\theta$; both coefficients are dimensionless.

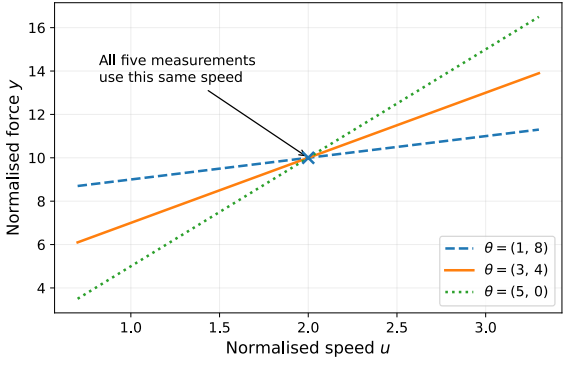
New synthetic example from now on:

$$\theta_0 = (3, 4)^T, \qquad y = 3u + 4.$$

The old data and the new synthetic data are different datasets.

Model and original data: [S0], pp. 32–35. New experiments and plots: calculations supplied with this lecture.

3. One speed reveals only one parameter combination



Take five exact measurements at $u = 2$:

$$X = \begin{bmatrix} 2 & 1 \\ \vdots & \vdots \\ 2 & 1 \end{bmatrix}_{5\times 2}, \qquad Y = \begin{bmatrix} 10 \\ \vdots \\ 10 \end{bmatrix}_{5\times 1}.$$

Every measurement says $2\theta_1 + \theta_2 = 10$. For every $a \in \mathbb{R}$,

$$\theta(a) = \begin{bmatrix} 3 + a \\ 4 - 2a \end{bmatrix} \quad \Rightarrow \quad X\theta(a) = Y.$$

The direction $h = (1, -2)^T$ is invisible: $Xh = 0$.

The experiment identifies $2\theta_1 + \theta_2$, not the two coefficients separately. Repeating this exact speed does not remove that ambiguity.

Scope: unrestricted linear parameters and this fixed experiment; not a general definition for every nonlinear dynamical model.

4. Identifiability for a fixed linear experiment

For $\theta \in \mathbb{R}^m$, call the parameters identifiable from X when $X\theta_a = X\theta_b$ implies $\theta_a = \theta_b$.

Criterion

Identifiability from $X \iff \ker X = \{0\} \iff \operatorname{rank} X = m \iff X^T X$ is positive definite.

Step 1: compare two candidates. Let $h = \theta_a - \theta_b$. Then

$$X\theta_a = X\theta_b \qquad \Longleftrightarrow \qquad Xh = 0.$$

Only $h = 0$ is allowed exactly when the nullspace is trivial. By rank–nullity, $\dim \ker X = m - \operatorname{rank} X$, so this is equivalent to $\operatorname{rank} X = m$.

Step 2: connect the criterion to the normal equations. For every h,

$$h^T(X^T X)h = (Xh)^T(Xh) = \|Xh\|_2^2 .$$

This is positive for every $h \neq 0$ exactly when $Xh \neq 0$ for every $h \neq 0$. Thus full column rank makes $X^T X$ invertible; rank deficiency makes it singular.

Linear-algebra setting: [S1], Ch. 5 and Ch. 12. Proof written out here; no noise distribution is needed for this rank criterion.

5. In this model, speed variation controls the rank

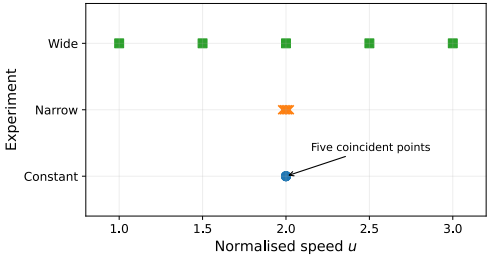
For rows $[u_k, 1]$, let $\bar{u} = N^{-1} \sum_k u_k$.

$$G = X^T X = \begin{bmatrix} \sum_k u_k^2 & \sum_k u_k \\ \sum_k u_k & N \end{bmatrix}.$$

Expand the determinant:

$$\begin{aligned} \det G &= N \sum_k u_k^2 - \Big(\sum_k u_k \Big)^2 \\ &= N \Big(\sum_k u_k^2 - N \bar{u}^2 \Big) \\ &= N \sum_k (u_k - \bar{u})^2. \end{aligned}$$

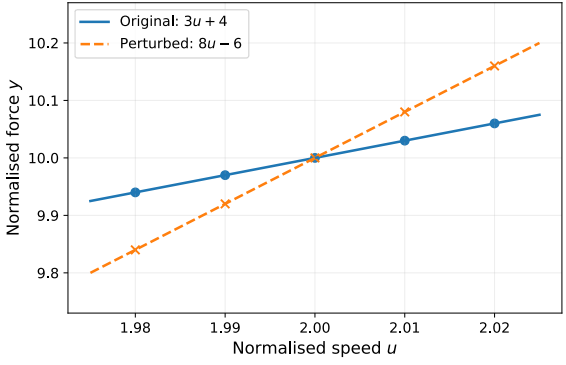
Expand each square and use $\sum_k u_k = N\bar{u}$ to check the last equality.



Design	N	$\sum (u_k - \bar{u})^2$
Constant	5	0
Narrow	5	0.001
Wide	5	2.5

At least two distinct speeds give full rank. A tiny nonzero spread gives uniqueness, but does not yet guarantee a reliable estimate.

6. A small change in the data can change the slope greatly



Use $u = (1.98, 1.99, 2, 2.01, 2.02)^T$. The exact data obey $y_k = 3u_k + 4$.
Construct a perturbation

$$\Delta y_k = 5(u_k - 2),$$

so its five values are
$(-0.10, -0.05, 0, 0.05, 0.10)$.
Then, point by point,

$$y_k + \Delta y_k = 3u_k + 4 + 5u_k - 10 = 8u_k - 6.$$

Hence the unique OLS estimate changes from $(3, 4)^T$ to $(8, -6)^T$.

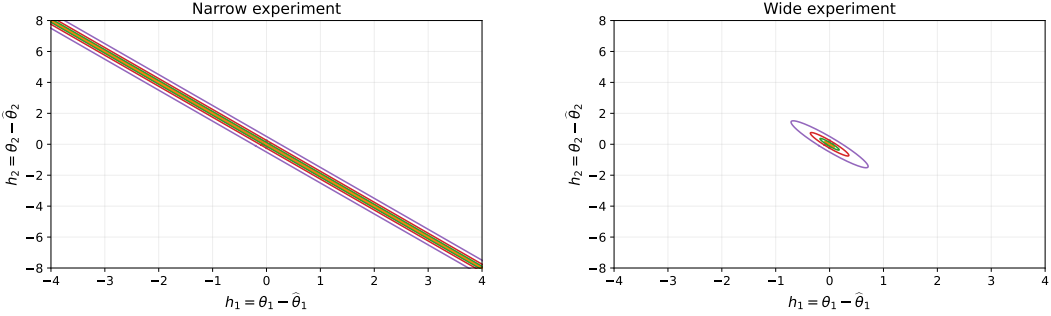
$$\frac{\|\Delta Y\|_2}{\|Y\|_2} \approx 0.00707 = 0.707\%.$$

This is a constructed sensitivity example, not a typical independent-noise sample. The negative friction coefficient is a warning about an unconstrained estimate, not a physical claim.

7. The loss has a long, narrow valley

At an OLS minimiser, $r = Y - X\widehat{\theta}_{\mathrm{LS}}$ satisfies $X^T r = 0$. Therefore

$$\|r - Xh\|_2^2 - \|r\|_2^2 = -2h^T X^T r + \|Xh\|_2^2 = \|Xh\|_2^2 .$$



Same coefficient-displacement axes and same loss-increase levels: $0.005, 0.02, 0.08, 0.32, 1.28$. Some narrow-design contours extend beyond the window.

A large coefficient change in a weak direction can have a small effect on the fitted data.

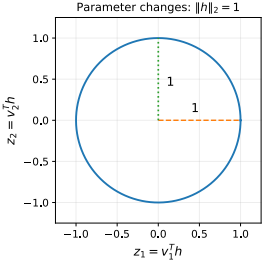
Quadratic-loss identity from the normal equations. The two plots use the stated synthetic design matrices.

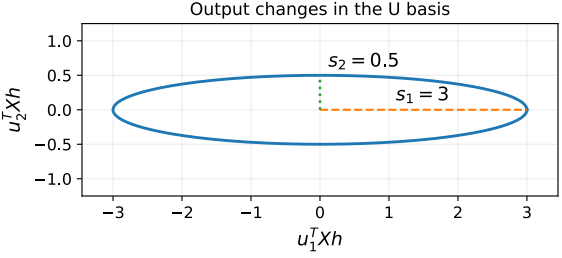
8. SVD separates the strong and weak directions

For $N \geq m$, the thin singular value decomposition is

$$X = U\Sigma V^T, \quad U^T U = I_m, \quad V^T V = I_m, \quad \Sigma = \mathrm{diag}(s_1, \ldots, s_m), \quad s_1 \geq \cdots \geq s_m \geq 0.$$

V has parameter directions v_j as columns; U has output directions u_j. The relation $Xv_j = s_j u_j$ means: one unit along v_j changes the output by s_j units.





Geometric example: $s_1 = 3$, $s_2 = 0.5$; these are not the friction dataset's singular values. Output axes are coordinates in the U basis, not two selected sensor readings.

SVD theorem: Gilbert Strang, MIT 18.06, Lecture 29 [S2]. Construction from $X^T X$ is in Appendix A1.

9. In SVD coordinates, least squares is scalar

Assume $s_m > 0$. Change coordinates, without changing lengths:

$$z = V^T\theta, \qquad b = U^T Y, \qquad Y_\perp = (I - UU^T)Y.$$

Then $Y = Ub + Y_\perp$, $U^T Y_\perp = 0$, and $X\theta = U\Sigma z$.

Step 1: split the residual into orthogonal parts.

$$\begin{aligned} \|Y - X\theta\|_2^2 &= \|U(b - \Sigma z) + Y_\perp\|_2^2 \\ &= \|b - \Sigma z\|_2^2 + \|Y_\perp\|_2^2 \\ &= \sum_{j=1}^{m} (b_j - s_j z_j)^2 + \|Y_\perp\|_2^2 . \end{aligned}$$

The cross term is zero because $U^T Y_\perp = 0$; $U^T U = I_m$ preserves the other norm.

Step 2: minimise each square. Its derivative is $-2s_j(b_j - s_j z_j)$, so

$$\widehat{z}_j = \frac{b_j}{s_j}, \qquad \widehat{\theta}_{\mathrm{LS}} = V\Sigma^{-1}U^T Y = \sum_{j=1}^{m} \frac{u_j^T Y}{s_j} v_j.$$

SVD [S2] plus the orthogonal decomposition of a least-squares residual [S1], Ch. 12.

10. Small singular values amplify uncertainty

A deterministic change in Y

Keep X fixed. From the SVD solution,

$$\Delta\widehat{\theta} = V\Sigma^{-1}U^T\Delta Y.$$

Use orthogonality; discarded squared components are nonnegative:

$$\begin{aligned}\left\|\Delta\widehat{\theta}\right\|_2^2 &= \sum_j \frac{(u_j^T\Delta Y)^2}{s_j^2} \\ &\le \frac{1}{s_m^2}\sum_j (u_j^T\Delta Y)^2 \\ &\le \frac{\|\Delta Y\|_2^2}{s_m^2}.\end{aligned}$$

Thus $\left\|\Delta\widehat{\theta}\right\|_2 \le \|\Delta Y\|_2 / s_m$. Equality is possible for ΔY along u_m.

Random measurement error

Let $\mathbb{E}\varepsilon = 0$ and $\mathrm{Cov}(\varepsilon) = \sigma^2 I_N$. For $z_{0j} = v_j^T\theta_0$,

$$\widehat{z}_j = z_{0j} + \frac{u_j^T\varepsilon}{s_j}.$$

The numerator has variance

$$u_j^T(\sigma^2 I_N)u_j = \sigma^2,$$

because $\|u_j\|_2 = 1$. Therefore

$$\boxed{\mathrm{Var}(\widehat{z}_j) = \frac{\sigma^2}{s_j^2}.}$$

This is the Gauss–Markov covariance formula, resolved into individual parameter directions.

Both results are derived here for a fixed, error-free X. No Gaussian distribution is required.

11. Condition number measures directional imbalance

For a full-column-rank X, define

$$\kappa_2(X) = \frac{s_1}{s_m}.$$

Indeed, if $h = Vz$, then $\|Xh\|_2^2 = \sum_j s_j^2 z_j^2$ and $\|h\|_2^2 = \sum_j z_j^2$. The largest and smallest gains are s_1 and s_m.

Experiment	Exact rank	s_1	s_m	$\kappa_2(X)$
Constant speed	1	5	0	∞
Narrow speed range	2	5.00008	0.01414	353.56
Wide speed range	2	5.19978	0.67994	7.65

Do not confuse two quantities. $1/s_m$ controls worst-case absolute amplification; s_1/s_m compares amplification across directions.

Multiplying X by a nonzero scalar changes both singular values, but not their ratio. A condition number alone does not specify the covariance or the noise level.

All numerical values computed from the three given matrices. Same fixed physical scaling in all comparisons.

12. A stable solver cannot create missing information

From $X = U\Sigma V^T$,

$$X^T X = V\Sigma U^T U\Sigma V^T = V\Sigma^2 V^T.$$

Its eigenvalues are s_j^2, so, at full column rank,

$$\boxed{\kappa_2(X^T X) = \kappa_2(X)^2.}$$

For the narrow experiment, $\kappa_2(X) \approx 353.56$ but $\kappa_2(X^T X) \approx 125008$.

Numerical issue

Forming the Gram matrix can magnify the effect of rounding. Solve least squares using QR or SVD rather than explicitly computing $(X^T X)^{-1}$.

Experimental issue

Even an exact solver gives $(8, -6)^T$ for our perturbed data. To address measurement sensitivity, improve the experiment or introduce justified additional assumptions.

LAPACK Users' Guide: QR-based and SVD-based least-squares solvers [S4]. Old rounding example retained in Appendix A6.

13. Ridge adds an explicit preference for smaller parameters

For the chosen dimensionless coordinates, define an *unaveraged* objective

$$J_\lambda(\theta) = \|Y - X\theta\|_2^2 + \lambda \|\theta\|_2^2, \qquad \lambda > 0.$$

The first term measures data mismatch; the second penalises parameter size.

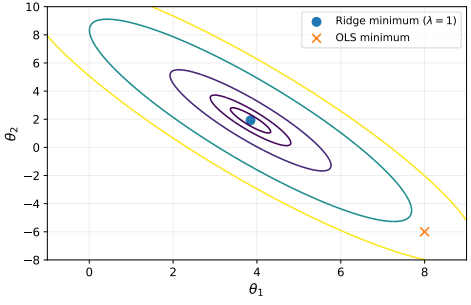
Find a stationary point.

$$\begin{aligned} J_\lambda(\theta) &= Y^T Y - 2\theta^T X^T Y \\ &\quad + \theta^T (X^T X + \lambda I_m)\theta, \\ \nabla J_\lambda(\theta) &= -2X^T Y + 2(X^T X + \lambda I_m)\theta. \end{aligned}$$

Set the gradient to zero:

$$(X^T X + \lambda I_m)\widehat{\theta}_\lambda = X^T Y.$$

We still need to show that the solution exists and is the unique global minimum.



Geometry only: $\lambda = 1$ here. The later numerical comparison uses $\lambda = 0.01$.

Quadratic regularised fitting: [S1], §15.4. Both normalised physical coefficients are deliberately penalised.

14. Ridge has a unique global minimum

Write $A = X^T X + \lambda I_m$. For every nonzero h,

$$h^T A h = \|Xh\|_2^2 + \lambda \|h\|_2^2 > 0.$$

The first term is nonnegative; the second is positive. Thus A is positive definite and invertible, even when X has deficient rank.

$$\boxed{\widehat{\theta}_\lambda = (X^T X + \lambda I_m)^{-1} X^T Y.}$$

Check global optimality, not just a zero gradient. Expand around $\widehat{\theta}_\lambda$:

$$\begin{aligned} J_\lambda(\widehat{\theta}_\lambda + h) - J_\lambda(\widehat{\theta}_\lambda) &= 2h^T(A\widehat{\theta}_\lambda - X^T Y) + h^T A h \\ &= 0 + h^T A h > 0 \qquad (h \neq 0). \end{aligned}$$

The first term vanishes by the stationary-point equation. Every other parameter vector has larger loss.

The penalised problem is now unique. This is not a proof that the original data distinguish every physical parameter.

All steps use $\lambda > 0$ and the full identity penalty in the stated coordinates.

15. Ridge changes the amplification in each SVD direction

Since $\|\theta\|_2^2 = \|z\|_2^2$, the variable part of the ridge loss is

$$\sum_j \left[(b_j - s_j z_j)^2 + \lambda z_j^2\right].$$

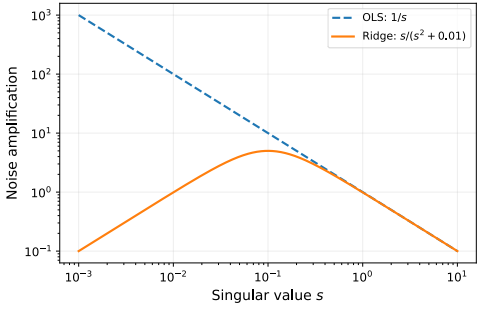
Differentiate its jth term:

$$-2s_j(b_j - s_j z_j) + 2\lambda z_j = 0.$$

Therefore

$$\widehat{z}_{\lambda j} = \frac{s_j}{s_j^2 + \lambda} b_j = \underbrace{\frac{s_j^2}{s_j^2 + \lambda}}_{q_j} \widehat{z}_{\mathrm{LS},j}$$

for $s_j > 0$. Here $0 < q_j < 1$.
If $s_j = 0$, the penalised solution sets $z_j = 0$: a preference, not information extracted from Y.



The plot shows observation-to-coefficient gain, not q_j. The relative shrinkage q_j is stronger when s_j is smaller.

SVD interpretation: [S3], §1.4. Shrinkage is directional; individual original coefficients need not change monotonically.